PAGE NO :
DATE :

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Assignment:

1] What is machine learning?

→ Machine Learning (ML) is a subset of Artificial Intelligence that enables computers to learn automatically from data & past experiences without being explicitly programmed.

2] Types of Machine Learning.

→ i] Supervised Learning
ii] Unsupervised Learning
iii] Reinforcement Learning.

3] Difference b/w AI & ML.

→ Artificial Intelligence is a board field that aims to create intelligent machines, while Machine Learning is a subset of AI that focuses on learning from data to improve performance automatically.

4] Lifecycle of Machine Learning.

→ i] Data Gathering
ii] Data Preparation
iii] Data Wrangling
iv] Data Analysis
v] Model Training
vi] Model Testing
vii] Deployment.

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PAGE NO :
DATE :

5] Structured Data.

⟹ Structured data is organized in a fixed format such as tables, databases & spreadsheet making it easy to analyz.

6] Unstructured Data.

⟹ Unstructured data does not follow a predefined format such as images, videos, & text data.

7] Supervised Learning.

⟹ It is a learning method where models are trained using labelled data to map input to output & make predictions.

8] Unsupervised Learning.

⟹ It is a method where models learn ~~problem~~ patterns from Unlabled data without supervision.

9] Model Selection.

⟹ It is the process of choosing the most appropriate algorithm that best fits the data & problem.

10] How to Train Data?

⟹ Training is done by feeding labeled data into the model so it can learn patterns, rules & relationships.

11] How to Test Data?

⟹ Testing involves evaluating the trained models using a seperate dataset to measure accuracy & performance.

12] Confusion Matrix.

⟹ It is a table used to evaluate classification performance by showing true positives, false positives, true negatives & false negatives.

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13] Correlation

⇒ It measures the relationship between variables & helps identify trends & dependendencies in data.

14] Importance of Regularization.

⇒ Regularization reduces model complexity, prevents overfitting & improves generalization.

15] Handling missing values.

⇒ Missing values can be handled by removing, replacing or imputing them using statistical methods.

16] Detecting ~~offer~~ outliners.

⇒ Outliners are extreme values that differ significantly & can be detected using statistical or visualization technique.

17] Advantages of Preprocessing.

⇒
- ✓ Improves data quality
- ✓ Removes noise & inconsistencies
- ✓ Enhances model accuracy.

18] Data sources.

⇒ Data can be collected from:
- ✓ Files
- ✓ Databases
- ✓ Internet
- ✓ Mobile devices

19] Feature Selection.

⇒ It is the process of selecting ~~proce~~ important variables for building efficient models.

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20] Monitering Data

→ Monitering involves checking model performance & updating it with new data to maintain accuracy.

21] Techniques in Supervised Learning

→ ✓ Regression
✓ Classification

22] Application of ML.

→ ✓ Image Recognition
✓ Speech Recognition
✓ Fraud Detection
✓ Self-driving cars.

23] History of ML.

→ ML evoled from early computational theories to mordern deep learning, with major milestones like IBM, Deep Blue & Alphago.

24] Evoluation of ML.

→ ML involed from rule-based systems to data-driven intelligent systems using neural networks, deep learning & real world application.

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